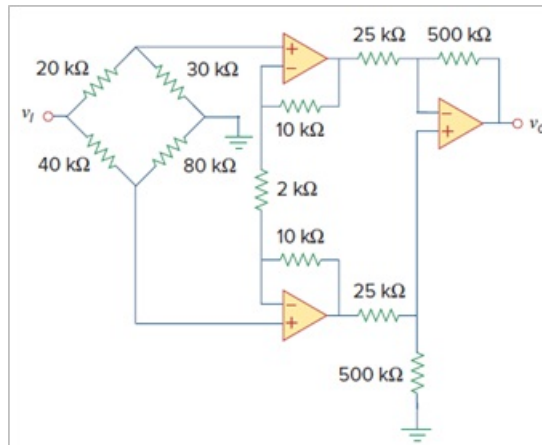


Problem

Figure 5.106 shows an instrumentation amplifier driven by a bridge. Obtain the gain v_o/v_i of the amplifier.

Figure 5.106:

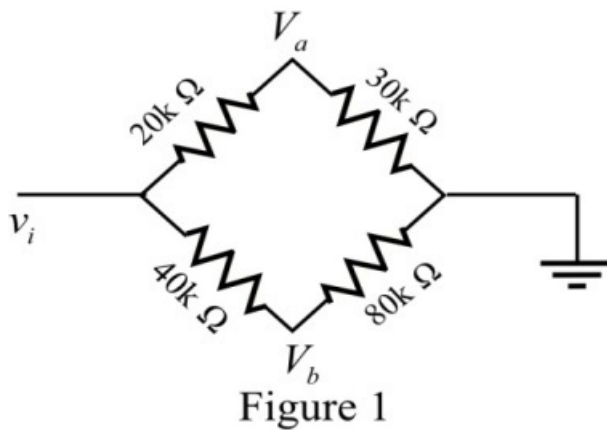


Step-by-step solution

Step 1 of 4

Refer to Figure 5.106 in the text book.

Consider the bridge circuit shown in Figure 1.



Step 2 of 4

Express the node voltages v_a and v_b in terms of the input voltage v_i .

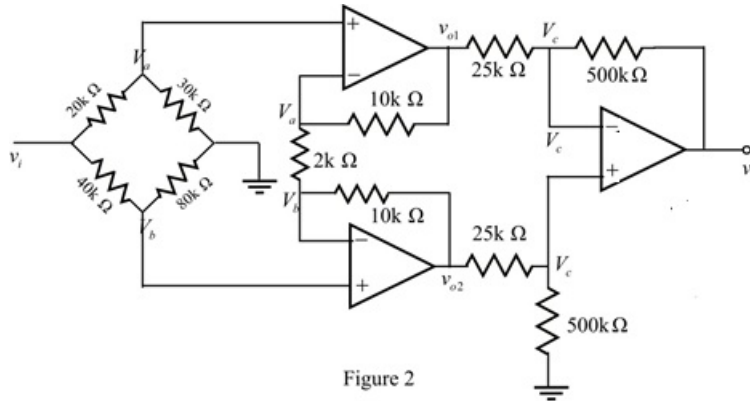
$$v_a = \left(\frac{30k}{30k + 20k} \right) v_i$$

$$= \frac{3}{5} v_i$$

$$v_b = \left(\frac{80k}{80k + 40k} \right) v_i$$

$$= \frac{2}{3} v_i$$

Indicate the node voltages in the circuit and draw the equivalent circuit.



Step 3 of 4

Apply Kirchhoff's voltage at node voltage v_a .

$$\frac{v_a - v_{o1}}{10k} + \frac{v_a - v_b}{2k} = 0$$

$$6v_a - v_{o1} - 5v_b = 0$$

$$6\left(\frac{3}{5}\right)v_i - v_{o1} - 5\left(\frac{2}{3}\right)v_i = 0$$

$$v_{o1} = \frac{4}{15}v_i$$

Apply Kirchhoff's voltage at node voltage v_b .

$$\frac{v_b - v_a}{2k} + \frac{v_b - v_{o2}}{10k} = 0$$

$$-5v_a + 6v_b - v_{o2} = 0$$

$$-5\left(\frac{3}{5}\right)v_i + 6\left(\frac{2}{3}\right)v_i = v_{o2}$$

$$v_{o2} = v_i$$

Step 4 of 4

Use voltage division principle to find the expression of v_c .

$$\begin{aligned} v_c &= \left(\frac{500k}{500k + 25k} \right) v_{o2} \\ &= \left(\frac{500}{525} \right) v_i \end{aligned}$$

Apply Kirchhoff's current law at inverting terminal of output op amp.

$$\begin{aligned} \frac{v_c - v_{o1}}{25k} + \frac{v_c - v_o}{500k} &= 0 \\ 21v_c - 20v_{o1} &= v_o \\ 21 \left(\frac{500}{525} \right) v_i - 20 \left(\frac{4}{15} \right) v_i &= v_o \\ 14.667v_i &= v_o \\ \frac{v_o}{v_i} &= 14.667 \end{aligned}$$

Thus, the gain $\frac{v_o}{v_i}$ is 14.667.